

PCS221 Cloud Computing – Microsoft Azure

Lab Assignment 5

VM Migration & Replication using Azure Site Recovery

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Course: PCS221 – Cloud Computing

Platform: Microsoft Azure

Existing VM Details (from Lab Assignment 4)

The same Ubuntu Apache Web Server VM created in Lab Assignment 4 was reused for this assignment.

VM Name: pcs221-webserver

OS: Ubuntu Server 22.04 LTS

Region: Central India

Public IP: 20.204.118.73

Apache Status: Running

Q1 – Configure Azure Site Recovery

Objective

Set up Azure Site Recovery to replicate the existing VM from Central India to South India for disaster recovery.

Step 1 – Create Recovery Services Vault

A new Recovery Services Vault was created in a different region from the source VM:

Vault Name: pcs221-recoveryvault

Region: South India

Resource Group: pcs221-rg-dr

Recovery Services Vault created successfully.

Step 2 – Enable Site Recovery Replication

Navigated to:

Recovery Services Vault -> Site Recovery -> Enable Replication

Source Settings

Source Type: Azure

Source Region: Central India

Source VM: pcs221-webserver

Target Settings

Target Region: South India

Target Resource Group: pcs221-failover-rg

Target Virtual Network: pcs221-dr-vnet

Target Availability: Single Instance

Replication initialization completed successfully.

Replication Status After Completion

VM Replication Status: Protected

Replication Health: Healthy

Last Sync Status: Successful

Protection State: Protected

The VM status showed Protected once replication was fully complete.

Q2 – Perform Test Failover

Objective

Verify that the replicated VM works correctly in the target region without affecting the live production VM.

Step 1 – Initiate Test Failover

Navigated to:

Recovery Services Vault -> Replicated Items -> pcs221-webserver -> Test Failover

Test Failover was configured with the following settings:

Recovery Point: Latest Processed

Virtual Network: pcs221-test-vnet

Test Failover initiated successfully.

Step 2 – Verify Test VM

A test VM was generated automatically with the following details:

VM Name: pcs221-webserver-test

Public IP: 20.219.84.126

Region: South India

SSH Connection

```
ssh azureuser@20.219.84.126
```

SSH connection was established successfully.

Apache Service Verification

```
sudo systemctl status apache2
```

apache2.service - The Apache HTTP Server

Active: active (running)

Webpage Verification

Opened the following URL in the browser:

<http://20.219.84.126>

Website loaded successfully on the Test Failover VM.

Step 3 – Cleanup Test Failover

After verification, the test failover was cleaned up:

Replicated Items -> pcs221-webserver -> Cleanup Test Failover

Cleanup completed successfully. Test VM resources were removed.

Q3 – Perform Actual VM Migration (Failover)

Objective

Perform a complete VM migration to South India using Azure Site Recovery failover and commit the migration permanently.

Step 1 – Initiate Failover

Navigated to:

Recovery Services Vault -> Replicated Items -> pcs221-webserver -> Failover

Failover was configured with:

Recovery Point: Latest

Shutdown VM Before Failover: Enabled

Failover started successfully.

Step 2 – Commit Migration

Once the failover completed, the migration was committed:

Replicated Items -> pcs221-webserver -> Commit

Migration committed successfully.

Migrated VM Details

Migrated VM Name: pcs221-webserver-dr

Region: South India

Public IP: 20.244.62.191

OS: Ubuntu Server 22.04 LTS

Verification After Migration

SSH Access

ssh azureuser@20.244.62.191

Connected to the migrated VM successfully.

Apache Service Check

sudo systemctl status apache2

apache2.service - The Apache HTTP Server
Active: active (running)

Website Access

http://20.244.62.191

Website loaded successfully after migration.

Data Consistency Check

- index.html present and accessible
- contact.html present and accessible
- Apache configuration intact
- Website content unchanged from original

Migration completed with zero data loss. All files and Apache configuration were intact in the target region.

GitHub Repository Submission

The following files were uploaded to the GitHub repository as per the assignment requirements:

- ip.txt – Public IP of migrated VM (20.244.62.191)
- before-migration.png – Screenshot of original VM webpage
- test-failover.png – Screenshot of Test Failover VM
- migrated-vm.png – Screenshot of Migrated VM

Conclusion

This lab gave hands-on experience with Azure Site Recovery for cross-region VM replication and migration. The VM was successfully replicated from Central India to South India. A test failover was run and verified before performing the actual migration. After committing the failover, the migrated VM was confirmed to be running Apache with all data intact and the webpage accessible via the new public IP 20.244.62.191. The exercise covered the full disaster recovery and migration workflow on Azure with zero data loss.

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